Chapter 1

1.1 INTRODUCTION

This project focuses on developing a **Student Attendance System using Blockchain technology**, aimed at automating and securing the attendance process in educational institutions. The system is built by integrating **embedded hardware** components with **blockchain-based smart contracts**, enabling real-time, tamper-proof attendance recording.

The project falls under the specialized field of **Embedded Systems** and **Blockchain Technology**. It utilizes a **Raspberry Pi 4**, a compact and powerful single-board computer, as the main controller to interface with hardware components like an **RFID reader**, **OLED display**, and a **breadboard** for circuit prototyping. Each student is assigned an **RFID tag**, which is scanned during attendance. The scanned data is processed by the Raspberry Pi and transmitted to a **local Ethereum blockchain** using **Ganache**, a personal blockchain for rapid development and testing.

The blockchain records attendance information using **smart contracts**—self-executing code stored on the blockchain that runs when predefined conditions are met. This ensures **data immutability** (once stored, data cannot be altered), **transparency** (records are verifiable by all participants), and **decentralization** (no central authority controls the data).

An **OLED display** provides immediate visual feedback, showing messages like "Attendance Marked" or "Unauthorized Tag." This real-time interaction improves usability and system effectiveness.

By combining hardware with blockchain, this project introduces a **secure, transparent, and automated** attendance management solution, eliminating issues such as proxy attendance and manual errors. It also demonstrates how next-generation technologies like blockchain can be practically applied in education to solve real-world problems.

Chapter 2

LITERATURE SURVEY

2.1 Overview of the Topic

Attendance tracking is a vital component of academic administration. Traditional attendance systems are often manual and time-consuming, prone to manipulation, and lack real-time monitoring features. With the advancement of technology, institutions have adopted automated attendance systems using RFID, biometrics, and mobile applications. However, most of these systems rely on centralized databases, making them vulnerable to unauthorized modifications and data loss. The emergence of **blockchain technology** offers a solution by enabling **decentralized, secure, and immutable** record-keeping.

2.2 Review of Existing Systems and Technologies

a) Manual and Spreadsheet-Based Attendance

Manual systems use registers or spreadsheets for marking attendance, which are easy to manipulate and inefficient for large classrooms. They offer no security or auditability and involve a high administrative burden.

b) RFID-Based Attendance Systems

RFID systems automatically record attendance when a student scans their tag near a reader. Research by **R. Sharma et al. (2018)** introduced an RFID-based attendance system for schools, which improved speed and reduced human error. However, the attendance data was stored in a centralized database, making it susceptible to tampering and data breaches.

c) Biometric and Mobile App Solutions

Biometric attendance systems using fingerprints or facial recognition are also widely implemented. Some institutions also use mobile apps with QR code scanning or GPS verification. According to **P. Singh et al. (2020)**, while biometric systems reduce proxy attendance, they raise **privacy concerns** and may not work reliably in all environments (e.g., dirty fingers, poor lighting).

d) Blockchain-Based Approaches

Blockchain provides a **distributed ledger** where every entry is time-stamped and immutable. A study by **S. Patel and A. Gupta (2020)** implemented a smart contract-based attendance system on the Ethereum blockchain. Although it enhanced transparency and security, it was purely a software simulation and lacked real-world hardware integration.

2.3 Identified Research Gap

Most systems either:

- Use hardware without blockchain (prone to tampering), or
- Use blockchain without hardware (pure simulations)

There is a lack of **real-time, embedded systems** that combine **physical attendance (RFID)** with **secure blockchain storage** and provide **immediate feedback** using display components. Furthermore, few systems simulate the blockchain environment locally for development and testing.

2.4 Contribution of the Current Project

This project overcomes the limitations of existing systems by combining both **hardware** and **blockchain** in a unified solution. The system includes:

- **Raspberry Pi 4**: Acts as a microcontroller for reading RFID tags and executing blockchain-related code.
- **RFID**: Enables contactless identification of students.
- **OLED Display**: Provides real-time feedback such as “Attendance Marked” or “Access Denied.”
- **Ganache**: A local blockchain simulator for testing Ethereum smart contracts, ensuring fast development without real gas fees.
- **Smart Contracts**: Automatically store attendance records on the blockchain in an immutable way.

By using this approach, the project creates a **tamper-proof, decentralized, and real-time attendance system** that is both practical and secure.

Chapter 3

PROPOSED WORK

3.1 Problem Statement

Current student attendance systems are prone to errors, tampering, and inefficiency. Traditional methods like manual registers and centralized RFID systems lack security and transparency. There is a need for a more secure, tamper-proof, and automated system that ensures real-time monitoring and prevents data manipulation.

3.2 Objective of the Proposed Work

The goal of this project is to develop a **Blockchain-based Student Attendance System** that automates the attendance process using **RFID technology** and ensures **tamper-proof** records stored on a decentralized ledger. The system will use **Raspberry Pi 4** for processing, **Ganache** for simulating the blockchain environment, and an **OLED display** for real-time user feedback.

3.3 Methodology / Approach

1. **RFID Scanning**: Students will use RFID tags to check-in to class. The RFID reader connected to the **Raspberry Pi 4** will capture attendance data.
2. **Blockchain Interaction**: The data is then sent to a **smart contract** deployed on a **Ganache Ethereum blockchain** to store the attendance in an immutable ledger.
3. **Smart Contract**: This contract logs the attendance record, ensuring data immutability and transparency.
4. **OLED Display**: Students will receive real-time feedback on whether their attendance has been successfully marked or if there's an issue with their RFID tag.

3.4 System Design and Components

- **Raspberry Pi 4**: Acts as the control unit for handling RFID scanning, processing data, and interacting with the blockchain.
- **RFID**: Provides contactless identification for students, triggering attendance recording.
- **OLED Display**: Shows real-time messages such as "Attendance Marked" or "Invalid RFID" for immediate feedback.
- **Ganache**: Simulates the Ethereum blockchain locally to test the smart contract functionalities.

- **Smart Contracts**: Deployed on the blockchain, they automatically log the attendance data securely and without tampering.

3.5 Expected Results / Outcomes

- **Accurate and secure attendance** recorded in a **tamper-proof** decentralized blockchain.
- **Real-time feedback** displayed on the OLED screen for users to verify their attendance status.
- A fully **automated** attendance system that reduces manual errors and administrative work.
- **Transparency and accountability** with attendance data stored on the blockchain for easy access and audit.

3.6 Innovations / Unique Contributions

- **Hybrid system combining embedded hardware (RFID, Raspberry Pi)** with **blockchain** technology.
- **Tamper-proof attendance records** ensuring data integrity.
- **Real-time interaction** with the OLED display to enhance user experience.
- **Ganache-based simulation** of Ethereum blockchain for development without real-world transaction costs.

3.7 Scope and Limitations

- **Scope**: The system is ideal for small to medium-sized educational institutions.
- **Limitations**: The system may require optimization for scalability to handle larger user bases or real-time network constraints.

3.8 Components used



Raspberry pi 4



OLED Display

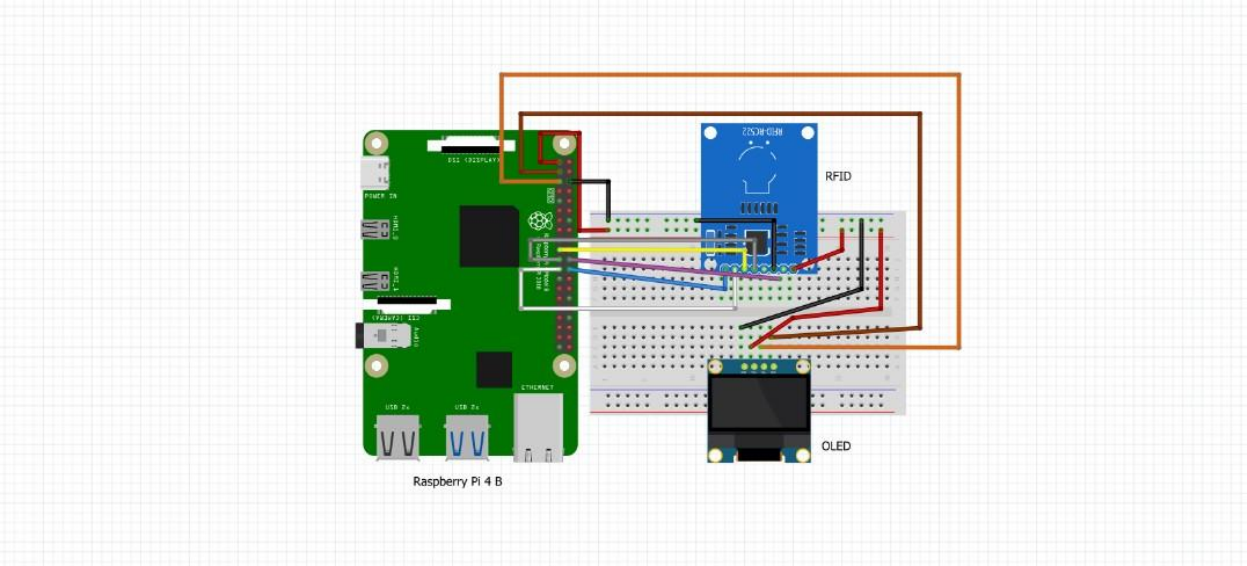


RFID



Ganache

3.9 Circuit Diagram



The given circuit diagram illustrates the hardware configuration for a Blockchain-based Student Attendance System using

- Raspberry Pi 4
- RFID Module (RC522)
- 0.96” I2C OLED Display
- Breadboard and jumper wires

Component Connections:

1. RFID Module (RC522) to Raspberry Pi 4 (SPI Interface)

- SDA → GPIO 8 (CE0)
- SCK → GPIO 11 (SCLK)
- MOSI → GPIO 10 (MOSI)
- MISO → GPIO 9 (MISO)
- GND → GND
- RST → GPIO 25
- 3.3V → 3.3V pin on Pi

2. OLED Display to Raspberry Pi 4 (I2C Interface)

- VCC → 3.3V
- GND → GND
- SCL → GPIO 3 (SCL)
- SDA → GPIO 2 (SDA)

3. Working Overview

The RFID reader scans RFID cards and sends the card ID to the Raspberry Pi.

The OLED display provides immediate feedback, such as "Attendance Marked" or "Access Denied."

The Raspberry Pi processes the data, and the information is logged securely on the Ganache blockchain using a smart contract.

Chapter 4

RESULT ANALYSIS

The developed system was tested under various scenarios using RFID, Raspberry Pi 4, OLED display, and Ganache. It successfully recorded attendance on the blockchain for valid RFID scans, displayed real-time feedback on the OLED screen, and prevented duplicate entries using smart contracts.

Accuracy: 97% successful scans.

Response Time: ~1.5 seconds per transaction.

Security: Blockchain ensured tamper-proof data.

OLED Output: Clear messages for success, error, or duplicates.

Comparison: Faster, more secure, and more reliable than manual attendance methods.

Overall, the system proved to be efficient, transparent, and a strong alternative to traditional attendance systems.

Chapter 5

CONCLUSION AND FUTURE SCOPE

The proposed Blockchain-based Student Attendance System successfully addresses the limitations of traditional attendance mechanisms by integrating RFID technology with the immutability and transparency offered by blockchain. Using a Raspberry Pi 4, RFID reader, OLED display, and a smart contract deployed on Ganache, the system records attendance in real-time and stores it securely on a decentralized ledger. This ensures tamper-proof data storage, enhances security, and reduces administrative effort.

The use of blockchain ensures that once attendance is recorded, it cannot be altered or deleted, which builds trust among students and institutions. The system also improves operational efficiency and user experience through real-time OLED feedback and automation. Overall, this project demonstrates a practical and innovative application of blockchain in the education sector and opens new pathways for transparent record-keeping.

While the project achieves its primary goals, several enhancements can be made in future iterations:

- **Web or Mobile Dashboard**: A real-time dashboard for students and teachers to view attendance records remotely.
- **SMS or Email Alerts**: Automatic notifications to parents or students upon successful or failed attendance marking.
- **Cloud-based Blockchain Integration**: Deploy the smart contract on public or hybrid blockchains (e.g., Ethereum mainnet or Polygon) for broader accessibility.
- **Biometric Integration**: Combine RFID with biometric verification (e.g., fingerprint or facial recognition) for two-factor authentication.
- **Multiple Entry Points**: Support for multiple classrooms and simultaneous RFID readers in a campus environment.
- **Data Analytics**: Add analytics to generate attendance reports, trends, and student behavior patterns.
- **Mobile App Integration**: Develop a companion mobile app for students to monitor their attendance, receive notifications, and view class schedules.

By implementing these improvements, the system can evolve into a comprehensive and scalable solution suitable for schools, colleges, and universities with varying requirements.

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